ENOXAPARIN

High-alert drug

Renal dose adjustment

CHEMICAL NAME Low-molecular-weight heparin (sodium salt, average MW ~4,500 Da) obtained by alkaline depolymerisation of porcine heparin benzyl ester

CLASS Low-molecular-weight heparin (LMWH) anticoagulant

TRADE NAMES Clexane · Lovenox · Inhixa · Enoxa

THERAPEUTIC USES / INDICATIONS

- VTE prophylaxis in medical and surgical patients, pregnancy.
- Treatment of DVT and PE.
- Acute coronary syndrome (NSTEMI, STEMI with thrombolysis/PCI).
- Bridging anticoagulation; extracorporeal circuits (haemodialysis).

ADMINISTRATION

- **Deep SC** into abdominal fold (2 inches from umbilicus), alternating sides; patient lying.
- Do not expel air bubble from prefilled syringe; do not rub site.
- Pinch skin fold throughout injection; insert needle fully at 90°.
- **IV** bolus in STEMI/PCI.

DOSAGES

- **Prophylaxis:** 40 mg SC once daily (20 mg if CrCl <30).
- **DVT/PE treatment:** 1 mg/kg SC every 12 h or 1.5 mg/kg once daily (CrCl <30: 1 mg/kg once daily).
- **NSTEMI:** 1 mg/kg every 12 h.
- **STEMI (<75 yr):** 30 mg IV bolus + 1 mg/kg SC every 12 h; elderly no bolus, 0.75 mg/kg.
- **Pregnancy:** weight-based (booking weight).

PHARMACOKINETICS

- **Onset:** SC 3–5 h peak anti-Xa. **Duration:** ~12 h.
- Bioavailability ~100% SC; predictable response (no routine monitoring).
- Excreted in urine — accumulates in renal impairment. t½: 4.5–7 h.

PHARMACODYNAMICS

- Binds **antithrombin** → mainly inhibits factor Xa (anti-Xa:anti-IIa ~3–4:1).
- Monitoring (if needed): anti-Xa level 4 h post-dose (renal failure, obesity, pregnancy, children).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site bruising, pain, haematoma.
- Minor bleeding.
- Raised liver enzymes.

ADVERSE EFFECTS (SERIOUS)

- Major bleeding.
- HIT (less than UFH).
- Spinal/epidural haematoma with neuraxial anaesthesia → paralysis.
- Hyperkalaemia; osteoporosis (long-term).

CONTRAINDICATIONS & CAUTIONS

- Active major bleeding; HIT history (within 100 days).
- Acute bacterial endocarditis; recent haemorrhagic stroke.
- Neuraxial anaesthesia timing: ≥12 h after prophylactic, ≥24 h after treatment dose before puncture/catheter removal.
- Severe thrombocytopenia.

KEY INTERACTIONS

- **Antiplatelets, NSAIDs, other anticoagulants:** ↑ bleeding.
- **ACE inhibitors, ARBs, K^+-sparing diuretics:** hyperkalaemia.

NURSING MANAGEMENT

- Weigh patient; check creatinine/CrCl before dosing.
- Correct technique; rotate sites; document site.
- Monitor platelets (baseline and periodically), Hb, K^+.
- Check timing with epidural/spinal procedures; neuro checks of legs.
- Watch for bleeding.

PATIENT EDUCATION

- Teach self-injection technique and sharps disposal.
- Bruising at the site is common; do not rub.
- Report bleeding, black stools, severe headache, leg weakness/numbness after spinal/epidural.
- Complete the full course (e.g. after surgery).

KEY POINTS

- LMWH; anti-Xa; SC, no routine monitoring.
- Reduce dose if CrCl <30.
- Prophylaxis 40 mg OD; treatment 1 mg/kg BD.
- Epidural timing rules; partially reversed by protamine.

SOURCES CHECKED **[414]** Inhixa 10,000 IU (100 mg) in 1.0 mL solution for injection in pre-filled syringe (UK SmPC) **[415]** AROVI 10,000 IU (100mg/1ml) pre-filled syringe with safety device (UK SmPC) **[416]** Enoxaparin Sodium, SC, IV (US label) **[417]** Enoxiluv Kit, SC, topical (US label)

Full references p. 432–457